```
cd /home/tplan/dc/elasticsearch

cat > instances.yml <<'EOF'
instances:

  - name: node-security-1
    dns:
      - node-security-1
      - es01
    ip:
      - 192.168.101.112

  - name: node-security-2
    dns:
      - node-security-2
      - es02
    ip:
      - 192.168.101.111

  - name: node-security-3
    dns:
      - node-security-3
      - es03
    ip:
      - 192.168.101.113

  - name: kibana
    dns:
      - kibana
      - localhost
    ip:
      - 192.168.101.111
EOF
```

3. ساخت CA با docker run

```
docker run --rm \
  -u 0 \
  -v /home/tplan/dc/elasticsearch/certs:/certs \
  docker.lib2.tiddev.com/elasticsearch/elasticsearch:9.2.2 \
  bash -c '
    set -e

    mkdir -p /certs

    bin/elasticsearch-certutil ca \
      --silent \
      --pem \
      --out /certs/ca.zip

    unzip -o /certs/ca.zip -d /certs

    rm -f /certs/ca.zip

    chmod -R 755 /certs
```

```
    echo
    echo "CA generated:"
    find /certs -type f
  '
```

بعد باید داشته باشی:

```
certs/
└── ca/
    ├── ca.crt
    └── ca.key
```

```
docker run --rm \
  -u 0 \
  -v /home/tplan/dc/elasticsearch/certs:/certs \
  -v /home/tplan/dc/elasticsearch/instances.yml:/instances.yml:ro \
  docker.lib2.tiddev.com/elasticsearch/elasticsearch:9.2.2 \
  bash -c '
    set -e

    bin/elasticsearch-certutil cert \
      --silent \
      --pem \
      --out /certs/certs.zip \
      --in /instances.yml \
      --ca-cert /certs/ca/ca.crt \
      --ca-key /certs/ca/ca.key

    unzip -o /certs/certs.zip -d /certs

    rm -f /certs/certs.zip

    chmod -R 755 /certs

    echo
    echo "Certificates generated:"
    find /certs -type f | sort
  '
```

باید خروجی چیزی شبیه این باشد:

```
certs/
├── ca/
│   ├── ca.crt
│   └── ca.key
│
├── node-security-1/
│   ├── node-security-1.crt
│   └── node-security-1.key
│
├── node-security-2/
│   ├── node-security-2.crt
│   └── node-security-2.key
│
├── node-security-3/
│   ├── node-security-3.crt
│   └── node-security-3.key
```

```
|
└── kibana/
    ├── kibana.crt
    └── kibana.key
```

5. چک certificateها

```
find /home/tplan/dc/elasticsearch/certs -type f | sort
```

```
openssl x509 \
  -in /home/tplan/dc/elasticsearch/certs/node-security-1/node-security-1.crt \
  -noout \
  -subject \
  -issuer \
  -ext subjectAltName
```

باید SAN شامل این باشد:

```
IP Address:192.168.101.112
```

6. حالا certs را منتقل کنیم

از ماشینی که certificateها را ساختی:

```
node 1 – .112
scp -r \
  /home/tplan/dc/elasticsearch/certs/ca \
  /home/tplan/dc/elasticsearch/certs/node-security-1 \
  tplan@192.168.101.112:/home/tplan/dc/elasticsearch/certs/
node 2 – .111
scp -r \
  /home/tplan/dc/elasticsearch/certs/ca \
  /home/tplan/dc/elasticsearch/certs/node-security-2 \
  tplan@192.168.101.111:/home/tplan/dc/elasticsearch/certs/
node 3 – .113
scp -r \
  /home/tplan/dc/elasticsearch/certs/ca \
  /home/tplan/dc/elasticsearch/certs/node-security-3 \
  tplan@192.168.101.113:/home/tplan/dc/elasticsearch/certs/
```

چون Kibana باشد 111. روی قرار است روی:

```
scp -r \
  /home/tplan/dc/elasticsearch/certs/kibana \
  tplan@192.168.101.111:/home/tplan/dc/elasticsearch/certs/
```

در نهایت

192.168.101.112 روی:

```
/home/tplan/dc/elasticsearch/certs/
├── ca/
└── node-security-1/
```

192.168.101.111 روی:

```
/home/tplan/dc/elasticsearch/certs/
├── ca/
├── node-security-2/
```

```
└── kibana/
```

192.168.101.113 روی:

```
/home/tplan/dc/elasticsearch/certs/
├── ca/
└── node-security-3/
```

1. روی هر سه سرور

مسیر:

```
cd /home/tplan/dc/elasticsearch
```

.env 192.168.101.112 روی

```
STACK_VERSION=9.2.2

NODE_NAME=node-security-1
NODE_IP=192.168.101.112

CLUSTER_NAME=elastic-cluster

ES_PORT=9200

ELASTIC_PASSWORD=ChangeMe123!
```

.env 192.168.101.111 روی

```
STACK_VERSION=9.2.2

NODE_NAME=node-security-2
NODE_IP=192.168.101.111

CLUSTER_NAME=elastic-cluster

ES_PORT=9200

ELASTIC_PASSWORD=ChangeMe123!
```

.env 192.168.101.113 روی

```
STACK_VERSION=9.2.2

NODE_NAME=node-security-3
NODE_IP=192.168.101.113

CLUSTER_NAME=elastic-cluster

ES_PORT=9200

ELASTIC_PASSWORD=ChangeMe123!
```

این فایل را عیناً روی هر سه سرور قرار بده:

```
version: "3.8"
```

```
services:
 elasticsearch:
   image: docker.lib2.tiddev.com/elasticsearch/elasticsearch:${STACK_VERSION}
   container_name: ${NODE_NAME}
   restart: unless-stopped
   network_mode: host
   environment:
    node.name: ${NODE_NAME}
    node.roles: master,data,ingest,transform
    cluster.name: ${CLUSTER_NAME}
    discovery.seed_hosts: >-
     192.168.101.111,
     192.168.101.112,
     192.168.101.113

   cluster.initial_master_nodes: >-
     node-security-1,
     node-security-2,
     node-security-3

   network.host: ${NODE_IP}

   http.port: 9200
   transport.port: 9300

   xpack.security.enabled: "true"

   xpack.security.http.ssl.enabled: "true"
   xpack.security.http.ssl.certificate: >-
     /usr/share/elasticsearch/config/certs/${NODE_NAME}/${NODE_NAME}.crt
   xpack.security.http.ssl.key: >-
     /usr/share/elasticsearch/config/certs/${NODE_NAME}/${NODE_NAME}.key
   xpack.security.http.ssl.certificate_authorities: >-
     /usr/share/elasticsearch/config/certs/ca/ca.crt

   xpack.security.transport.ssl.enabled: "true"

   xpack.security.transport.ssl.verification_mode: certificate

   xpack.security.transport.ssl.certificate: >-
     /usr/share/elasticsearch/config/certs/${NODE_NAME}/${NODE_NAME}.crt

   xpack.security.transport.ssl.key: >-
     /usr/share/elasticsearch/config/certs/${NODE_NAME}/${NODE_NAME}.key

   xpack.security.transport.ssl.certificate_authorities: >-
     /usr/share/elasticsearch/config/certs/ca/ca.crt

   ELASTIC_PASSWORD: ${ELASTIC_PASSWORD}
   ES_JAVA_OPTS: "-Xms1g -Xmx1g"
 mem_limit: 2g
 ulimits:
   memlock:
     soft: -1
     hard: -1
 volumes:
   - esdata:/usr/share/elasticsearch/data
```

```
  - ./certs:/usr/share/elasticsearch/config/certs:ro

healthcheck:
  test:
    [
      "CMD-SHELL",
      "curl --fail --silent --show-error --cacert
/usr/share/elasticsearch/config/certs/ca/ca.crt -u elastic:$${ELASTIC_PASSWORD}
https://localhost:9200/_cluster/health > /dev/null"
    ]
  interval: 10s
  timeout: 10s
  retries: 30
  start_period: 90s

volumes:

 esdata:
   name: elasticsearch_esdata
```

5. قبل از بالا آوردن Elasticsearch

خیلی مهم: روی هر سه سرور:

```
sudo sysctl -w vm.max_map_count=262144
```

برای دائمی شدن:

```
echo "vm.max_map_count=262144" | sudo tee /etc/sysctl.d/99-elasticsearch.conf
sudo sysctl --system

docker compose up -d

curl \
  --cacert /home/tplan/dc/elasticsearch/certs/ca/ca.crt \
  -u 'elastic:ChangeMe123!' \
  https://192.168.101.111:9200
```

Kibana

```
STACK_VERSION=9.2.2

NODE_NAME=node-security-2
NODE_IP=192.168.101.111

CLUSTER_NAME=elastic-cluster

ELASTIC_PASSWORD=ChangeMe123!

KIBANA_SYSTEM_PASSWORD=KibanaSystem123!

ENCRYPTED_SAVED_OBJECTS_KEY=9f6c8e2b1a7d4c5e9f6a8b3c2d1e7f4a9b6c3d8e1f5a2c7b9d4e6f8a1
c3e5b7
SECURITY_ENCRYPTION_KEY=4e8a2c6f1b9d3e7a5c8f2b6d9e1a4c7f3b8d5e2a6c9f1b4d7e8a3c5f6b9d
REPORTING_ENCRYPTION_KEY=7b3e9f1c5a8d2e6f4b9c1a7d3e5f8b2c6d9a4e1f7b3c5d8e2a6f9b1c4d7e
```

ولی یک نکته مهم درباره kibana_system

پسورد kibana_system را نباید صرفاً با KIBANA_SYSTEM_PASSWORD در environment
Elasticsearch انتظار داشته باشیم که ساخته شود.

پسورد kibana_system، باید بعد از بالا آمدن اولین Elasticsearch با API تنظیم کنیم:

```
curl \
  --cacert certs/ca/ca.crt \
  -u "elastic:ChangeMe123!" \
  -X POST \
  "https://192.168.101.111:9200/_security/user/kibana_system/_password" \
  -H "Content-Type: application/json" \
  -d '{
    "password": "KibanaSystem123!"
  }'
```

```
  certs/
├── ca/
│   └── ca.crt
│
└── kibana/
    ├── kibana.crt
    └── kibana.key
```

```
services:

  kibana:
    image: docker.lib2.tiddev.com/kibana/kibana:${STACK_VERSION}

    container_name: kibana

    restart: unless-stopped

    network_mode: host

    environment:

      # ======================================================
      # Kibana Server
      # ======================================================

      SERVER_NAME: kibana
      SERVER_HOST: 192.168.101.111
      SERVER_PORT: 5601

      # ======================================================
      # Elasticsearch
      # ======================================================

      ELASTICSEARCH_HOSTS:
'["https://192.168.101.111:9200","https://192.168.101.112:9200","https://192.168.101.
113:9200"]'
```

```
      ELASTICSEARCH_USERNAME: kibana_system
      ELASTICSEARCH_PASSWORD: ${KIBANA_SYSTEM_PASSWORD}

      ELASTICSEARCH_SSL_CERTIFICATEAUTHORITIES:
/usr/share/kibana/config/certs/ca/ca.crt

      # ============================================================
      # Kibana HTTPS
      # ============================================================

      SERVER_SSL_ENABLED: "true"

      SERVER_SSL_CERTIFICATE: /usr/share/kibana/config/certs/kibana/kibana.crt

      SERVER_SSL_KEY: /usr/share/kibana/config/certs/kibana/kibana.key

      # ============================================================
      # Encryption Keys
      # ============================================================

      XPACK_ENCRYPTEDSAVEDOBJECTS_ENCRYPTIONKEY: ${ENCRYPTED_SAVED_OBJECTS_KEY}

      XPACK_SECURITY_ENCRYPTIONKEY: ${SECURITY_ENCRYPTION_KEY}

      XPACK_REPORTING_ENCRYPTIONKEY: ${REPORTING_ENCRYPTION_KEY}
      MONITORING_UI_CCS_ENABLED: "false"
    volumes:

      - ./certs:/usr/share/kibana/config/certs:ro
```